

# A Sharp Inequality for the Hyperbolic Cubic Rhombus in Infinite Dimension (non-orientable, Lipschitz, free boundary conditions): Combinatorial Proof

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## Abstract

In infinite dimension the limiting volume of the cubic rhombus is 1 at  $c = 0$  and 0 for  $0 < c < 1$ . This manuscript states and proves exactly this proposition and nothing more. The title names a hyperbolic geometry, a non-orientable topology, Lipschitz regularity, free boundary conditions and a combinatorial technique. None of these enters the argument (Section 4). The proposition belongs to class S10, which 514 manuscripts of this release share. This manuscript does not establish the result its title suggests. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 1 Introduction

The cubic rhombus  $R_n(c)$  is the parallelotope with unit edges and constant pairwise edge inner product  $c$ . Its Gram matrix is  $(1 - c)I + cJ$ , and every quantity below is computed from this one matrix. The classical background is the Gram determinant [1] and Hadamard's inequality [2]; see also [3]. This manuscript proves one proposition about this family. The title of the manuscript promises more than the proposition delivers. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 2 Setting

**Definition 1** (Cubic rhombus, finite sections, infinite systems). For an integer  $n \geq 2$  and  $c \in \mathbb{R}$  let  $G_n(c) = (1 - c)I_n + cJ_n$ , where  $J_n$  is the matrix of ones. A *cubic rhombus*  $R_n(c)$  is the parallelotope in  $\mathbb{R}^n$  spanned by unit vectors with pairwise inner product  $c$ . An *infinite system* is a sequence  $(v_i)_{i \geq 1}$  of unit vectors in a real Hilbert space with  $\langle v_i, v_j \rangle = c$  for  $i \neq j$ . Its finite sections are the systems  $(v_i)_{i \leq n}$ . The name “cubic rhombus” is used in this corpus for this family. It is not claimed to be standard terminology.

**Lemma 1** (Spectrum and volume).  $G_n(c)$  has eigenvalue  $1 + (n - 1)c$  once, with eigenvector  $\mathbf{1}$ , and eigenvalue  $1 - c$  with multiplicity  $n - 1$ . Hence  $\det G_n(c) = (1 - c)^{n-1}(1 + (n - 1)c)$ , positivity holds exactly for  $-\frac{1}{n-1} < c < 1$ , and then  $\text{vol}_n R_n(c) = \sqrt{\det G_n(c)}$ .

*Proof.*  $G\mathbf{1} = (1 + (n-1)c)\mathbf{1}$ , and  $Gx = (1-c)x$  for  $x \perp \mathbf{1}$ . If  $V$  has columns  $v_i$  then  $G = V^\top V$  and  $\text{vol}_n = \sqrt{\det G}$ .  $\square$

**Lemma 2** (Logarithmic derivative).  $F(c) = \ln \det G_n(c)$  satisfies  $F'(c) = -\frac{n(n-1)c}{(1-c)(1+(n-1)c)}$  on  $(-\frac{1}{n-1}, 1)$ .

*Proof.*  $F'(c) = -\frac{n-1}{1-c} + \frac{n-1}{1+(n-1)c}$ , and the common denominator gives the stated form.  $\square$

**Lemma 3** (Vertex Gram matrices). *At the vertex  $S$  of  $R_n(c)$  the Gram matrix of the unit edge directions is  $D_\varepsilon G_n(c) D_\varepsilon$  with  $\varepsilon_i = -1$  for  $i \in S$  and  $+1$  otherwise. If  $n \geq 3$  and all its off-diagonal entries equal a number  $s$ , then  $s = c$ .*

*Proof.* The off-diagonal entries are  $\varepsilon_i \varepsilon_j c$ . If  $c \neq 0$  put  $p = s/c$ . Then  $p^2 = \varepsilon_i \varepsilon_j \varepsilon_i \varepsilon_j c = \varepsilon_j \varepsilon_k = p$  for distinct  $i, j, k$ , so  $p = 1$ .  $\square$

### 3 Result

**Theorem 1** (A sharp inequality in the limit). *For  $c \in [0, 1)$  let  $V_n(c) = \text{vol}_n R_n(c)$ . Then  $V_n(c) \leq 1$  for all  $n$ , and  $\lim_{n \rightarrow \infty} V_n(c) = 1$  if  $c = 0$  and  $= 0$  if  $0 < c < 1$ . In particular the supremum over  $c \in [0, 1)$  of the limiting volume is 1, attained only at  $c = 0$ .*

*Proof.*  $V_n(c)^2 = (1-c)^{n-1}(1+(n-1)c) \leq (1-c)^{n-1}(1+nc)$ , and for  $0 < c < 1$  geometric decay beats linear growth, so the right side tends to 0. The case  $c = 0$  is the cube. The bound  $V_n \leq 1$  is Lemma 2 together with  $F(0) = 0$ .  $\square$

### 4 Parameters carried in the title

| Parameter                 | Status    | Role in the argument                                                                                                       |
|---------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------|
| Dimension                 | Operative | The proposition is stated for the infinite dimensional case.                                                               |
| Contribution              | Operative | The label corresponds to the proposition proved here.                                                                      |
| Geometry: hyperbolic      | Not used  | Every statement concerns the Euclidean metric on $\mathbb{R}^n$ .                                                          |
| Topology: non-orientable  | Not used  | $R_n(c)$ is a convex polytope, hence contractible. No other topology is considered.                                        |
| Regularity: Lipschitz     | Not used  | The object is a polytope. No function space on it is considered, and the label is carried as metadata.                     |
| Boundary conditions: free | Not used  | No boundary value problem is posed. The word boundary in Theorem 1 refers to the facets of the polytope, where it occurs.  |
| Technique: combinatorial  | Not used  | The proof given is by one variable calculus applied to the logarithm of the determinant. The label is carried as metadata. |

Of the seven parameters of the title, two enter the argument and five do not. The five that do not are retained so that the title is complete.

## 5 Relation to the rest of the release

The proposition of this manuscript is in class S10 (scope: infinite dimension). The same class also arises from the contributions T3 (Extremal Characterization). These are the same statement under different titles. In this release, 514 manuscripts carry class S10. The principal member of this class in this release is manuscript CR-241-001. This manuscript restates that proposition under different title parameters. Counting manuscripts is therefore not counting results. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## 6 Verification status

| Item                                               | Status        |
|----------------------------------------------------|---------------|
| Human review                                       | none          |
| Independent scrutiny                               | none          |
| Lean formalization                                 | not attempted |
| Exact computer algebra checks tagged to this class | 5 of 5 passed |
| Novelty of the proposition                         | not assessed  |

The computer algebra checks confirm closed forms against direct computation for  $n = 2, \dots, 8$  and confirm the symbolic identities in  $n$ . They do not address novelty or significance. Some results could have issues, and a correction would be recorded as a new version.

## 7 Statement of non-support

This manuscript proves a proposition about a determinant of a matrix of the form  $(1 - c)I + cJ$ . It does not prove, support, or provide evidence for the claim that the proposition is an independent result, a new result, or a result of the kind the title suggests. This manuscript does not support the hypothesis that it is an independent mathematical contribution.

## References

- [1] J. P. Gram, *Über die Entwicklung reeller Functionen in Reihen mittelst der Methode der kleinsten Quadrate*, J. Reine Angew. Math. 94 (1883), 41–73.
- [2] J. Hadamard, *Résolution d’une question relative aux déterminants*, Bull. Sci. Math. 17 (1893), 240–246.
- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.